

STAT GR5293 · SPRING 2026

Communication Protocol Effects on Multi-Agent System Efficiency and Task Completion

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Project Scope

PROBLEM

Why communication protocol is an uncontrolled design choice in multi-agent LLM systems.

CONTRIBUTION

A controlled benchmark, reusable code, logged artifacts, and a protocol-selection demo.

EVALUATION

360 pipeline runs with token, latency, and completion metrics across three domains.

The Communication Format Gap

Multi-agent LLM systems are increasingly built as pipelines: agents plan, execute, critique, retrieve, or integrate intermediate work.

Most frameworks expose inter-agent communication as a flexible engineering choice, but do not answer a basic empirical question.



Frameworks make the communication format flexible, but rarely tell us which format is efficient or reliable.

Three Guiding Questions

1

How much does protocol choice affect token cost and latency?

2

How much does protocol choice affect task completion quality?

3

Does the best protocol change across math, reading, and news?

Fixed Three-Agent Pipeline



HELD CONSTANT

Model, temperature, agent roles, system prompts, and max tokens by domain.

MANIPULATED

The inter-agent communication protocol only.

Four Communication Protocols

NL

Plain English prose with no structured formatting.

```
plain text
```

Markdown

Headings and bullets for readable intermediate reasoning.

```
# Heading
```

JSON

Validated JSON object with one parse-retry.

```
{ }
```

Shared Memory

A blackboard snapshot injected into every downstream agent.

```
state.json
```

4 × 3 Factorial Design

4

protocols

NL · Markdown · JSON ·

Shared Memory

3

domains

GSM8K · SQuAD · curated

news

360

pipeline runs

10 samples × 3 reps per

cell

EFFICIENCY

Prompt tokens · completion tokens · total
tokens · wall-clock latency

QUALITY

Numeric exact match · SQuAD token-F1 ·
ROUGE-based news scoring

What We Built

01

A reusable protocol benchmark for a fixed three-agent LLM pipeline.

02

A 360-run dataset with 1,080 message-level logs for audit and error analysis.

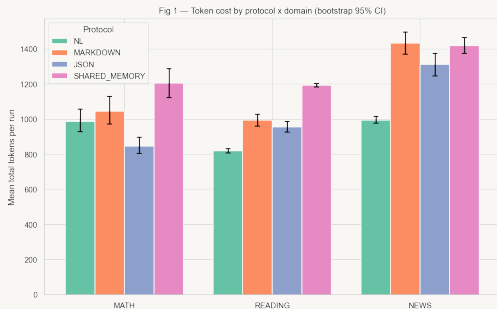
03

Statistical analysis: ANOVA, Tukey HSD, effect sizes, and bootstrap intervals.

04

A Streamlit demo that turns offline results into protocol recommendations.

Token Cost by Protocol and Domain

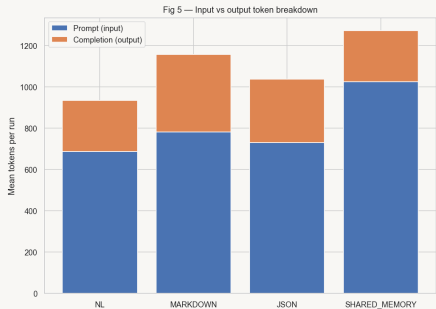


$$\eta^2 = .262$$

protocol effect size

Shared memory is the most expensive protocol in every domain, averaging +338 tokens per run versus NL.

Prompt vs. Completion Tokens

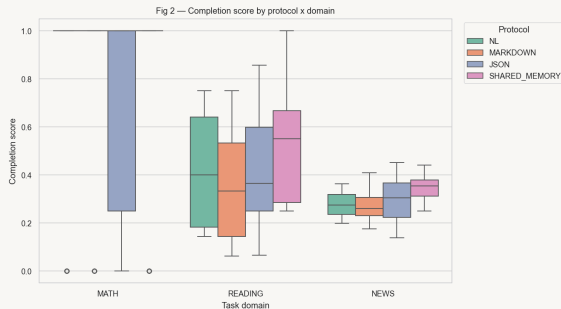


Input

token overhead

Shared Memory increases prompt tokens because each agent reads the accumulated blackboard. Markdown increases completion tokens because headings and bullets add verbosity.

Completion Quality by Domain

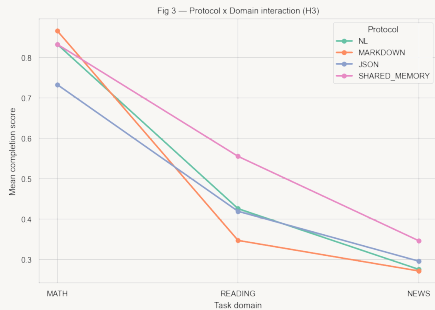


.017
math effect size

.102
reading effect size

.197
news effect size

Protocol × Domain Interaction



The best protocol changes across domains.

Math protocols cluster closely; reading and news reward the richer shared-memory context. Pooled completion interaction is not significant at $\alpha = .05$, but token interaction is strong.

Efficiency vs. Effectiveness Trade-off

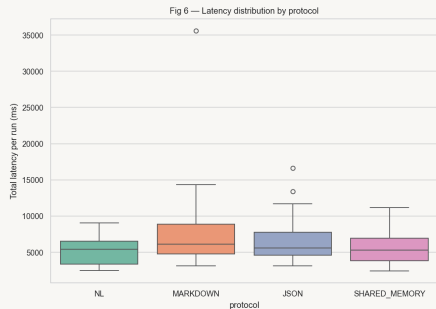


Math: choose Markdown for best completion, JSON for lowest cost.

Reading: choose Shared Memory for quality; NL for budget.

News: choose Shared Memory when factual completeness matters.

Latency Distribution by Protocol



$$F = 16.55$$

$$p < .001, \eta^2 = .070$$

Domain complexity dominates latency variance. Protocol still has a measurable operational effect, and token-efficient protocols tend to reduce estimated API cost.

Protocol Selection Demo

1

User enters a free-text task.

2

The app classifies the task domain.

3

It selects the empirically best protocol from
`results_summary.csv`.

4

The three-agent pipeline runs and reports tokens, latency, cost, and final answer.

Artifact: single-page Streamlit demo in `app.py`, with manual protocol override and raw message logs.

Limitations and Future Work

LIMITATIONS

- Only 10 samples per cell.
- Three repetitions per condition.
- All agents use `gpt-4o-mini`.
- Three-agent chain only.

FUTURE WORK

- Larger domain samples.
- XML, YAML, and hybrid protocols.
- Heterogeneous agent models.
- Dynamic protocol switching.

Choose Protocol by Domain and Priority

Do not choose an agent communication protocol globally. Choose it by domain and by whether the priority is cost or completion quality.

COST

NL and JSON are strong budget choices.

QUALITY

Shared Memory helps reading and news.

ARTIFACT

Code, results, figures, and demo are reproducible.

Cell Summary (N = 30 per cell)

Protocol	Domain	Tokens	Prompt	Completion	Score
NL	Math	988	665	323	.833
NL	Reading	819	717	102	.426
NL	News	995	680	315	.275
Markdown	Math	1046	695	351	.867
Markdown	Reading	995	794	201	.347
Markdown	News	1433	860	574	.271
JSON	Math	847	600	247	.733
JSON	Reading	955	777	178	.419
JSON	News	1313	815	498	.296
Shared Memory	Math	1204	931	273	.833
Shared Memory	Reading	1192	1095	97	.556
Shared Memory	News	1418	1052	366	.346